File Systems: FAT and NTFS

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Section- A

File Systems

A file system is a method employed by operating systems to store, organise, and facilitate access to data on storage devices. It acts as the backbone for how information is managed on your computer.

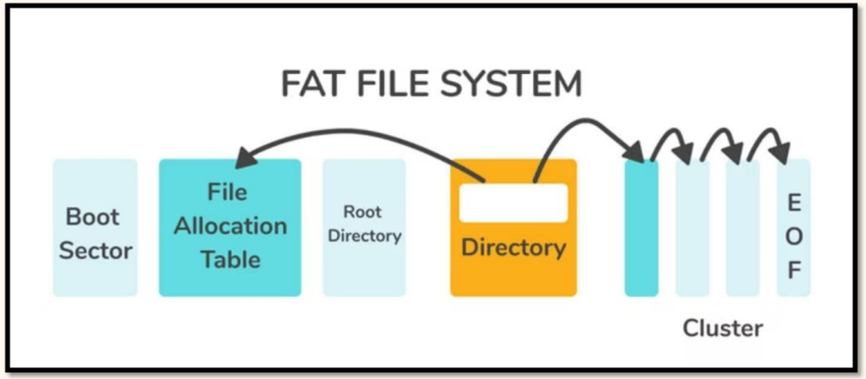
Core Functionality

File systems assign names to files, store critical metadata (like creation date and size), and efficiently manage available storage space.

Key Examples

- FAT (File Allocation Table)
- NTFS (New Technology File System)
- exFAT (Extended File Allocation Table)
- ext3/ext4 (Third/Fourth Extended File System)

Introduction to FAT (File Allocation Table)



The File Allocation Table (FAT) is one of the earliest file systems, developed by Microsoft in 1977. It's known for its simplicity and widespread compatibility.

Widespread Use

FAT remains common in:

- **USB drives**
- **SD cards**
- **Older floppy disks**
- **Cross-platform data exchange**

Key Variants

Over the years, FAT has evolved:

- **FAT12 (earliest, for small volumes)**
- **FAT16 (improved, larger volumes)**
- **FAT32 (most common, larger volumes and files)**

FAT Architecture: Cluster Chains and Fragmentation

1

Cluster Allocation

Files are stored in one or more clusters. If a file is larger than a single cluster, it's broken into pieces and stored in a chain of clusters.

2

Directory Entries

Each directory entry contains crucial file information: filename, timestamps (creation, modification), and the starting cluster of the file.

3

Non-Contiguous Storage

A key characteristic is that a file's clusters do not need to be contiguous on the disk. This can lead to fragmentation but allows for flexible storage.

FAT Variants: Capacity Limitations

Each FAT variant has specific limitations regarding the maximum volume size and maximum file size it can handle. Understanding these limits is crucial for choosing the right file system for a given storage device.

Type	Max Volume Size	Max File Size
FAT12	32 MB	32 MB
FAT16	2 GB (DOS)	2 GB
FAT32	2 TB (Windows)	4 GB

FAT32's 4 GB file size limit is a significant constraint for modern applications dealing with large media files or databases.

FAT: Advantages and Disadvantages

While FAT offers broad compatibility, its simplicity comes with certain trade-offs, particularly in modern computing environments.

Advantages

- **Simple and easy to implement, reducing overhead.**
- **Universally supported by nearly all operating systems (Windows, macOS, Linux).**
- **Excellent for removable drives due to its broad compatibility.**

Disadvantages

- **No built-in file security or permissions, making it less suitable for shared or sensitive data.**
- **Maximum file size limit (4 GB for FAT32) restricts its use for large files.**
- **No journaling feature, increasing the risk of data loss or corruption during system crashes.**

Numerical Example: FAT3 2 Internal Fragmentation

Let's illustrate how storage is allocated and potential waste (internal fragmentation) occurs within a FAT32 system.

> **Question: A file of 18 KB is to be stored in a FAT32 system. The cluster size is 4 KB.**
>
> **Find:**
>
> - **Number of clusters used**
> - **Internal fragmentation (wasted space)**

FAT Numerical Solution (Step-by-Step)

Step 1 : Clusters Needed

$$\frac{18\ \text{KB}}{4\ \text{KB}} = 4.5 \Rightarrow$$

Round up to 5 Clusters

Since clusters are allocated as whole units, 4.5 clusters means 5 clusters are required.

Step 2 : Total Allocated Size

5 Clusters × 4 KB/Cluster = 20 KB

This is the total disk space reserved for the 18 KB file.

Step 3 : Internal Fragmentation

20 KB (Allocated) − 18 KB (File Size) = 2 KB

This 2 KB is wasted space within the last allocated cluster, demonstrating internal fragmentation.

Summary of FAT Numerical Example

This table summarises the calculation, clearly showing how internal fragmentation arises in a FAT32 system when a file does not perfectly fill its allocated clusters.

Parameter	Value
File Size	18 KB
Cluster Size	4 KB
Clusters Used	5
Allocated Space	20 KB
Internal Fragmentation	2 KB

Introduction to NTFS (New Technology File System)

NTFS is Microsoft's advanced file system, designed for robustness, security, and efficiency in modern operating systems, particularly Windows NT and its successors.

Enhanced Capabilities

NTFS supports significantly larger files and volumes compared to FAT, along with rich metadata capabilities.

Advanced Security

Key features include Access Control Lists (ACLs), allowing granular permissions for files and folders.

Data Integrity

Journaling ensures file system consistency, even after power failures or system crashes, by logging changes before applying them.

NTFS Architecture

The core of NTFS is the Master File Table (MFT), a comprehensive database that stores all metadata for every file and directory on the volume.

File Compression

Efficiently reduces file sizes to maximise available disk space.

File Encryption (EFS)

Provides robust, integrated file-level security for sensitive data.

Disk Quotas

Enables administrators to set limits on disk space usage per user.

Journaling

Logs all changes to ensure data integrity and prevent corruption.

FAT vs NTFS: A Feature Comparison

Understanding the key differences between FAT32 and NTFS is essential for choosing the appropriate file system for various storage needs and operating environments.

Feature	FAT32	NTFS
Max File Size	4 GB	~16 EB (Exabytes)
Security	None	Yes (Access Control Lists)
Journaling	No	Yes
OS Compatibility	Universal	Mostly Windows
File Compression	No	Yes

While FAT32 offers broader compatibility, NTFS excels in modern environments with its superior capacity, security features, and data integrity.

Where are FAT and NTFS Used?

FAT

- USB drives
- SD cards
- Legacy devices

NTFS

- Windows internal storage
- Servers
- Secure enterprise systems

Conclusion

Our exploration of file systems highlights the distinct characteristics and applications of FAT and NTFS, alongside the practical implications of their designs.

FAT: Simple, Yet Outdated

FAT's straightforward design ensures broad compatibility, making it ideal for smaller, removable media. However, its lack of modern features and security limitations render it unsuitable for large, critical systems.

NTFS: Robust & Secure

NTFS stands out with its advanced capabilities, including journaling for data integrity, comprehensive security through Access Control Lists (ACLs), and efficient management of metadata for enhanced reliability.

Fragmentation Inefficiency

Numerical illustrations underscore a key drawback of FAT: internal fragmentation. This inherent inefficiency leads to wasted disk space, especially noticeable with smaller files on larger clusters.

Thank You!!